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# Current Developments and Challenges in the Field of Biohybrid Neural Interfaces—A Scoping Review

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**Biohybrid neural interfaces (BNIs) will be pivotal for the future of advancing neuroprosthetics and neuromodulation technologies, though they have not yet been widely adopted into routine practice. This scoping review aims to consolidate recent advancements, identify ongoing challenges, and examine methodologies in biohybrid neural interfaces to highlight knowledge gaps and inform future research. Following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews (PRISMA-ScR) framework, electronic literature searches of Medline/Embase, IEEE Xplore, Scopus, Web of Science, and the Cochrane Library, covering studies from 2000 onwards, focus on peer reviewed articles in English. A total of 25 articles are included in this scoping review, comprising 15 primary research studies, 7 review articles, and 3 related reports. These studies explore the integration of living cells, induced pluripotent stem cells (iPSCs), neurotrophic factors, hydrogels, and conductive polymers to enhance neural function and biocompatibility. Despite encouraging advances, challenges such as immune response, device durability, and signal fidelity remain critical barriers to clinical translation. Despite advancements, the field faces challenges that require further research. Future efforts should focus on refining bioactive coatings, advancing biohybrid designs, and developing next-generation neural recording systems to improve long-term integration and performance.**

## 1. Introduction

Implantable neural interfaces (NIs) have been widely used to treat or assist individuals with sensory, motor, or other neural functional impairments, including epilepsy, depression, Parkinson's disease, and deafness.[1,2] Brain–computer interfaces (BCIs) represent a subset of neural interface technologies that enable direct communication pathways between the brain and external devices, facilitating cognitive monitoring through the recording and interpretation of brain activity to decode users' motor intentions, cognitive states, and neurological conditions.[3] However, implantable neural interfaces face significant challenges due to limited biocompatibility, which can lead to foreign body reactions, tissue rejection, mechanical instability, and progressive loss of local neurons.[4–6] For invasive neural interfaces that require direct contact with target neuronal tissue, such as intracortical microelectrode arrays used in motor BCIs, both scar tissue and gliosis form around the electrodes as part of the post-implantation inflammatory response.[7] In contrast, for interfaces that do not directly contact neuronal tissue, such as cochlear implants, fibrotic tissue formation may still occur following initial surgical trauma. Additionally, the inherent anatomical gap between the implant and target neurons in such systems can result in imprecise stimulation and cross-channel interference.[8,9] These biocompatibility issues often lead to

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**DOI: 10.1002/aelm.202500259**

reduced functionality over time, ultimately necessitating surgical replacement procedures that pose additional risks, including infection.[10]

Biohybrid neural interfaces (BNIs) offer a promising approach to bridging the gap between electronic devices and biological systems.[11,12] As defined in recent consensus efforts, BNIs represent neural interface systems that incorporate living biological components, typically cells or tissues, as integral functional elements that mediate the interaction between electronic devices and the nervous system.[13] By combining neural interface engineering principles with cell transplantation methodologies, BNIs utilize living cells as biocompatible intermediaries, offering a more biologically integrated solution compared to conventional synthetic interfaces.[14] One fundamental BNI approach involves coating conventional electrodes with a biocompatible hydrogel matrix containing viable neurons prior to implantation. This strategy offers several potential advantages over traditional neural interfaces. First, the additional hydrogel layer increases the mechanical compliance of the electrode assembly, reducing insertion trauma and minimizing the acute inflammatory response.[15] Second, neurites extending from the transplanted neurons can establish functional synaptic connections with host neural tissue, enhancing implant-tissue integration and fundamentally altering the mechanism by which implants communicate with target neuronal populations.[16] This biological integration may improve long-term implant functionality, enhance signal transmission fidelity, and mitigate chronic foreign body responses. Furthermore, the extended neurite networks from BNIs enable more spatially precise signal transmission to specific neuronal subpopulations, potentially addressing the anatomical gap limitations encountered in applications such as cochlear implantation.[17]

A more specialized subtype of BNIs is known as living electrodes, in which the implanted biological component serves not merely as a biocompatible coating, but as the primary input/output unit for signal recording and stimulation. In these constructs, information exchange is mediated directly by living cells rather than synthetic electronics, offering a radically different paradigm for neuromodulation and prosthetic integration.[13] Building on these principles, recent biohybrid strategies have demonstrated that integrating living neuronal or muscle-derived constructs into neural devices can promote functional regeneration, synaptic connectivity, and biological signal amplification, highlighting their translational relevance for disorders such as peripheral nerve injury and Parkinson's disease.[13,18]

Despite the significant therapeutic potential of biohybrid neural interfaces, this field remains in early developmental stages. Current understanding of technological innovations, implementation challenges, and optimal methodological approaches requires systematic evaluation and synthesis.[13,18] This scoping review aims to comprehensively analyze current developments, identify key challenges, and examine the diverse methodological approaches within the biohybrid neural interface literature, while also exploring advancements in related neural prosthetic technologies. By focusing on the intersection of biohybrid systems, neural interfacing technologies, biomaterials science, and neural tissue engineering, this review examines how these integrated approaches can restore neural functions, improve long-term tissue integration, promote neurobiological compatibility, and address fundamental limitations in current neural prosthetic systems.

The primary objective of this scoping review is to elucidate current developments and challenges in biohybrid neural interfaces, with particular emphasis on technological innovations, methodological diversity, and clinical application outcomes.

## 2. Methods and Analysis

### 2.1. Protocol and Registration

The present scoping review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews (PRISMA-ScR).[19] The protocol was registered on Open Science Framework on September 10, 2024 (osf.io/s467c).

### 2.2. Eligibility Criteria

The inclusion criteria were as follows: 1) peer-reviewed research articles, 2) review articles, 3) meta-analyses, 4) conference papers, and 5) doctoral theses, all published in English. The exclusion criteria were: 1) non-peer-reviewed articles, 2) editorials, 3) articles not published in English; and 4) studies not directly related to BNIs.

### 2.3. Information Sources and Search Strategy

The search strategy was developed by a clinical librarian (VP) and peer-reviewed by two librarian colleagues using the Peer Review of Electronic Search Strategies checklist.[20] It was also evaluated in accordance with the PRISMA-S guidelines.

This search strategy focused on identifying relevant publications from 2000 onwards across six key databases: Medline (via Ovid), Embase (via Ovid), Cochrane Library, Scopus, Web of Science (Core Collection), and IEEE Explore. The search terms included "neuronal interface*", "neural interface*", "neural cell implant*", "hydrogel neuronal coating", "neural integrat*", "neural signal transmi*", "biohybrid*", "cell-seeded", "bioelectro*",

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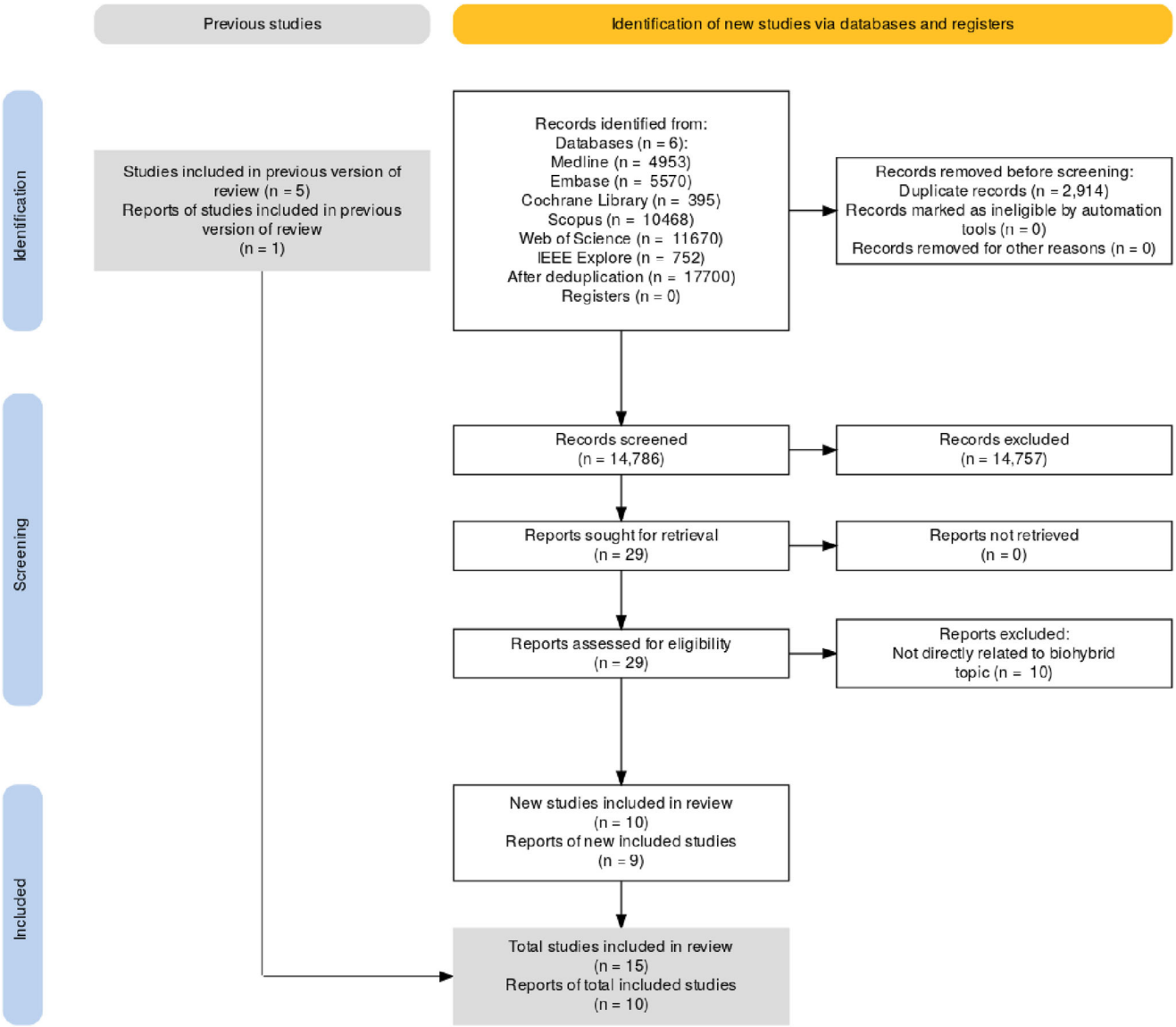


**Figure 1.** PRISMA-ScR flow diagram of study selection process. The figure outlines the procedures and eligibility criteria applied in selecting articles for inclusion in the review.

"bio-electro*", "microtissue*", "biocompatible", "living", "electrode*", "interface*", "implant*", "prosthe*", "probe*", and "microelectrode*". These terms were designed to capture studies related to neural interfaces, biocompatibility, and bioelectronic systems, with a focus on innovative implants and prosthetics. The full search strategy for each database is described in supplemental data 1.

### 2.4. Studies Selection

The search was conducted on July 17, 2024. Retrieved references were exported and uploaded to Rayyan (https://www.rayyan.ai/; Rayyan, Cambridge, MA, USA), where duplicates were removed. A pilot screening of 50 randomly selected titles and abstracts was performed to refine the screening criteria and ensure consistency in the application. Two independent reviewers (YL and JM) screened all titles and abstracts against the eligibility criteria, classifying articles as include, exclude, or maybe. Discrepancies were resolved through discussion.

Following the title and abstract screening, full-text screening was conducted to ensure consistent application of the eligibility criteria. The same two independent reviewers subsequently assessed the full texts of all articles marked for inclusion during the initial screening phase.

If consensus could not be reached during the title and abstract screening, a third senior reviewer provided the final decision. Articles marked as maybe were included in the full-text review to allow for a more detailed assessment. Any disagreements regarding full-text inclusion were resolved through discussion among reviewers, with the third reviewer intervening when necessary. For full-text articles with unclear or missing eligibility informa-


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tion, attempts were made to contact the study authors for clarification. If no clarification was obtained, the studies were excluded due to insufficient information, and this was documented accordingly.

In addition to the database searches, the reference lists of included articles were reviewed to identify relevant studies that might have been missed during the initial search. These studies were then screened, assessed, and subsequently included in the final review.

### 2.5. Data Extraction

To ensure consistency in data collection, a detailed data extraction template was developed. The template specified the information to be extracted from each included study, including the type of research, author names, year of publication, journal, the challenges associated with conventional neural interfaces that the study aimed to address, and the specific developments and challenges related to biohybrid neural interfaces (e.g., fabrication, application, or future prospects). For original research articles, additional details were recorded, such as population characteristics or models used, interventions employed, the type and species of interfacing cells, and the type of carrier scaffold utilized. The template was collaboratively developed by the research team to ensure comprehensive coverage of all relevant aspects of the reviewed studies.

### 2.6. Synthesis and Presentation of Results

Preliminary data checks were performed to ensure consistency and accuracy by cross-verifying the extracted data against the original sources. Any discrepancies or missing entries were resolved through discussion within the data extraction team, and all corrections were systematically documented. A PRISMA-ScR flow diagram (**Figure 1**) was used to illustrate the review process. The final synthesis mapped current developments and challenges in biohybrid neural interfaces, and was presented using tables, figures, and diagrams to enhance clarity and impact.

## 3. Results

### 3.1. Study Selection

From an initial 17700 studies, 2914 duplicates were removed. After screening 14786 records, 29 articles were retrieved for eligibility assessment, of which 10 were excluded for not being directly related to the biohybrid topic. An additional 6 relevant studies were identified through citation tracking of the included articles, comprising 5 original research articles and 1 review article. Ultimately, the review included 15 primary studies (**Table 1**), along with 3 reports and 7 review articles (**Table 2**). The screening process is summarized in Figure 1.

### 3.2. Characteristics of Included Studies

Fifteen primary research studies focused on the development of biohybrid systems, addressing key aspects such as device design, material optimization, and fabrication techniques. In addition, ten review articles and related reports provided broader perspectives on the field. For instance, one review traced the historical development and applications of biohybrid devices across the central nervous system, peripheral nervous system, and sensory systems,[14] while another offered a comprehensive overview of recent advancements and future directions in biohybrid neural interfaces.[21]

Six of these reviews explored the key milestones, challenges, and strategic approaches in the development of bioactive neural interfaces. In addition, several works focused on specific technical aspects, including electrode material properties, biological surface coatings, and strategies for cellular integration. For instance, the use of organic materials such as conductive polymers, carbon nanotubes, and hydrogels for enhancing electrode biocompatibility.[22] Surface modifications to improve neural tissue integration have also been discussed,[17] alongside foundational bioengineering principles and design frameworks for constructing biohybrid neural systems.[15] Furthermore, two reports presented longitudinal data from two-year and four-year follow-ups assessing the safety and functional performance of biohybrid cochlear electrodes.[23,24]

### 3.3. Findings from the Selected Publications

#### *3.3.1. Challenges in Conventional NIs Targeted by BNIs*

*Insertion Trauma*: The initial implantation of electrodes causes acute mechanical damage, including compression, plasma extravasation, and disruption of neurons and glial cells. This trauma can trigger an immune response that accelerates chronic inflammation.[16,25,26]

*Foreign Body Response, Chronic Inflammation, and Scar Formation*: Following the initial insertion trauma, a significant challenge identified across most of the included studies is the foreign body response elicited by conventional neural implants. This response leads to chronic inflammation, glial scar formation, and reduced device performance due to increased impedance.[1,15–17,21,22,25–35]

*Mechanical Mismatch*: The mechanical properties of conventional implant materials, such as silicon or platinum, often do not match the soft nature of neural tissue. This mismatch can lead to continuous microtrauma, contributing to inflammation, scar formation, and reduced long-term implant performance.[1,15,21,22,26,27,29–31,33,36,37]

*Limited Specificity in Neuromodulation*: Many conventional implants lack the specificity needed to target individual neuronal subtypes or specific neural compartments. This limitation reduces the precision of neuromodulation and often results in broad side effects.[21,32,35,37]

#### *3.3.2. Other Challenges Mentioned by Individual Studies*

*Low Charge Injection Capacity*: Limitations of conventional metallic electrodes, particularly their low charge injection capacity that reduces the efficiency of electrical stimulation, have been noted in prior reviews.[22]


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**Table 1.** Primary research studies on biohybrid neural interfaces (2002–2023).

| Type of research | Year | Author(s) | Carrier scaffold | Interfacing cells | Population characteristics | Interventions |
|---|---|---|---|---|---|---|
| Primary peer-reviewed study | 2002 | Stieglitz et al.[12] | Autologous femoral vein containment | Fragmented fetal spinal cord | Rat | Sciatica nerve implantation |
| Primary peer-reviewed study | 2009 | Purcell et al.[28] | Alginate hydrogel scaffold | Embryonic day 14 (E14) murine cortical neural stem cells | Rat | Cortical implantation |
| Primary peer-reviewed study | 2010 | Azemi et al.[38] | Laminin-coated silicon surface | Neural progenitor cells derived from the subependymal zone of rats | Rat | Cortical implantation |
| Primary peer-reviewed study | 2011 | Richter et al.[27] | Fibrin hydrogel | Pancreatic stem cells derived from adult rats | Rat | Cortical implantation |
| Primary peer-reviewed study | 2011 | Wise et al.[39] | Alginate capsule | Cells isolated from choroid plexus of neonatal pigs | Neonatal deafened cat | Cochlear implantation |
| Primary peer-reviewed study | 2013 | Green et al.[29] | Poly(vinyl alcohol) (PVA) and sericin-based degradable hydrogel | Neural cells (PC12) | In vitro study | None (in vitro study) |
| Primary peer-reviewed study | 2014 | Faveri et al.[26] | Fibrin hydrogel | Hippocampal neurons from E18 rat embryos or glial cells from postnatal day zero (P0) rat pups | Rat | Cortical implantation |
| Primary peer-reviewed study | 2014 | Tang-Schomer et al.[30] | Poly-L-lysine on silk film with patterned microgrooves | Primary rat cortical cells (E18) | Mouse | Brain surface implantation |
| Primary peer-reviewed study | 2016 | Roemer et al.[25] | Fibrin adhesive coating on silicone-based cochlear implants | Bone marrow-derived mononuclear cells | Human patients of hearing loss | Cochlear implantation |
| Primary peer-reviewed study | 2017 | Goding et al.[31] | Phenol-modified PVA with gelatin and sericin | Mixture of primary glial cells from postnatal day 1 (P1) and neural progenitor cells (NPCs) from gestational day 15 (GD15) Sprague Dawley rat fetuses | None | In vitro culture |
| Conference paper | 2020 | Paul and Hindges[37] | Not mentioned (direct culture on the microneedles) | E18 primary hippocampal rat neurons | In vitro culture | In vitro culture |
| Primary peer-reviewed study | 2021 | Adewole et al.[32] | Microcolumn scaffolds fabricated with agarose filled with ECM (mixture of rat tail collagen and mouse laminin) | Neurons from genetically engineered lines expressing GFP/mCherry (from E18 rats) | Sprague-Dawley rats | Cortical implantation |
| Conference paper | 2023 | Ruff[34] | Matrigel-based target structure for axonal growth and transition | Retinal neurons seeded onto the microelectrode array | In vitro study | In vitro study |
| Primary peer-reviewed study | 2023 | Ahmed et al.[33] | Gelatin methacrylate hydrogel, polymer microneedles, silicon base chip with gold-coated microwells | Cortical neurons from pre-natal (E18) C57BL/6 mice | Rat cadaver implantation to test mechanical stability | Intracortical implantation |

(*Continued*)

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**Table 1.** (Continued)

| Type of research | Year | Author(s) | Carrier scaffold | Interfacing cells | Population characteristics | Interventions |
|---|---|---|---|---|---|---|
| Primary peer-reviewed study | 2023 | Rochford et al.[35] | Fibrin hydrogel deposited on parylene-based microelectrode arrays (MEAs) | Human iPSC-derived skeletal myocytes | Hsd:RH-Foxn1rnu athymic nude rats | Ulnar and median nerve implantation |

### *3.3.3. Developments in Biohybrid Neural Interfaces*

Biohybrid systems are advancing prosthetic development by integrating living biological cells with synthetic scaffolds or electrodes to enhance neural integration, reduce foreign body response, and improve long-term functionality. Key innovations include the use of novel materials, advanced fabrication techniques, and biologically enhanced electrodes.

Stieglitz et al. proposed a polyimide-based sieve electrode with ring structures designed to facilitate axonal regrowth through sieve holes, demonstrating potential for muscle reinnervation in rat sciatic nerve models.[12] Similarly, Purcell et al. developed parylene-C-based neural probes seeded with neural stem cells (NSCs) encapsulated in alginate hydrogel, which improved neuronal density and reduced glial encapsulation in rats.[28] Azemi et al. utilized laminin coatings to enhance neural progenitor cell adhesion on silicon probes, promoting cell differentiation, survival, and migration in vivo.[38]

Richter et al. used stem cell coatings on polyimide films to mimic the native neural environment, demonstrating reduced astrocytic responses and improved long-term integration.[27] In the auditory system, Wise et al. encapsulated choroid plexus cells in alginate membranes, combining neurotrophin delivery with electrical stimulation to improve spiral ganglion neuron (SGN) survival in neonatal deafened cats.[39]

Hydrogel-based approaches have also shown promise. Green et al. introduced conductive hydrogel-coated poly(3,4-ethylenedioxythiophene) (PEDOT) and polyvinyl alcohol (PVA) electrodes, which enhanced charge storage capacity and reduced glial scarring.[29] Goding et al. developed a dual-layer hydrogel electrode that combined a non-degradable conductive layer (PEDOT) with a degradable phenol-modified PVA layer mixed with gelatine and sericin, improving mechanical compliance and reducing chronic inflammation.[31] De Faveri et al. utilized fibrin hydrogel-coated microelectrodes to reduce astrocytic responses and improve long-term electrophysiological performance in rats.[26] Tang-Schomer et al. employed silk-based biomaterials for compartmentalized neural circuits, achieving minimal inflammation and stable integration in mice.[30] Silk films promoted cell viability and glial alignment, supporting their

**Table 2.** Review articles and reports on biohybrid neural interfaces (2014–2023).

| Author(s) | Year of publication | Title of the study | Journal name |
|---|---|---|---|
| Aregueta-Robles et al.[22] | 2014 | Organic Electrode Coatings for Next-Generation Neural Interfaces. | Front Neuroeng, 2014. 7: p. 15. |
| Thompson et al.[1] | 2016 | Regenerative Electrode Interfaces for Neural Prostheses. | Tissue Eng Part B Rev, 2016. 22(2): p. 125–35. |
| Goding et al.[17] | 2017 | Living Bioelectronics: Strategies for Developing an Effective Long-Term Implant with Functional Neural Connections | Adv Funct Mater, 2017. 28(12). |
| Mitovska Fenov et al.[23] | 2018 | Two-year results with the biohybrid cochlear electrodes:Safe electrode? | Laryngo-Rhino-Otologie, 2018. 97(S 02). |
| Serruya et al.[21] | 2018 | Engineered Axonal Tracts as "Living Electrodes" for Synaptic-Based Modulation of Neural Circuitry | Adv Funct Mater, 2018. 28(12). |
| Adewole et al.[36] | 2019 | Bioactive Neuroelectronic Interfaces | Front Neurosci, 2019. 13: p. 269. |
| Rochford et al.[14] | 2020 | When Bio Meets Technology: Biohybrid Neural Interfaces | Adv Mater, 2020. 32(15): p. e1903182. |
| Mitovska Fenov et al.[24] | 2021 | Four-year results with the biohybrid cochlear electrodes: Safe electrode? | Laryngo-Rhino-Otologie, 2021. 100(S 02). |
| Shi et al.[16] | 2021 | Bio-integrative Design of The Neural Tissue-Device Interface | Curr Opin Biotechnol, 2021. 72: p. 54–61. |
| Boulingre et al.[15] | 2023 | Biohybrid neural interfaces: improving the biological integration of neural implants | Chem Commun (Camb), 2023. 59(100): p. 14745–14758. |


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suitability for long-term brain–machine interfaces. Roemer et al. applied bone marrow-derived mononuclear cells in fibrin-coated cochlear implants, demonstrating long-term outcomes comparable to those of conventional implants.[25] Follow-up studies confirmed their stability and safety of this approach.[23,24]

Recent innovations include high-aspect-ratio microneedles for neuron culture,[37] which support cell growth and synapse formation, demonstrating potential for enhanced neurostimulation precision with reduced trauma and immune response. Adewole et al. introduced optogenetically enabled microtissue-engineered neural networks that formed long axonal tracts, allowing simultaneous stimulation and monitoring.[32] Ahmed et al. developed biohybrid microelectrode arrays incorporating gelatin methacrylate hydrogel, exhibiting mechanical flexibility, stability, and suitability for high-resolution neural recordings based on preliminary in vitro findings.[33]

Further conceptual advancements include retinal neuron-based artificial optic nerves[34] and induced pluripotent stem cell (iPSC)-derived skeletal myocytes integrated with microelectrodes for motor-driven prostheses, achieving successful chronic implantation.[32] These efforts underscore the potential of biohybrid neural interfaces to enhance implant functionality, biocompatibility, and long-term integration.

#### 3.3.4. Current Challenges in Developing BNIs

The development of BNIs presents a range of technical and biological challenges, reflecting the complexity of integrating living tissue with synthetic devices. The included studies highlight key issues encountered during fabrication, application, and future prospects, which are summarized below.

*Immune Response and Glial Encapsulation*: Although biohybrid neural interfaces can reduce reactive gliosis and scar formation, host immune reactions and rejection of implanted cells remain significant challenges. Purcell et al. observed increased glial encapsulation due to scaffold degradation and exposure to immune factors.[28] Similarly, immune response and graft rejection have been identified as key barriers, alongside the difficulty of maintaining neuronal survival in a hostile or degenerative environments.[21] Graft rejection is particularly concerning for BNIs using standardized allogeneic constructs.[36] Wise et al. noted that fibrous tissue formation could limit the efficacy of neurotrophin delivery in cochlear implants,[39] while Goding et al. reported chronic foreign body responses that reduced cell survival.[31]

*Scaffold, Interface, and Hydrogel Stability*: Hydrogel scaffolds, which are integral to many BNIs, face significant challenges in balancing degradation rates with long-term structural stability. De Faveri et al. and Green et al. highlighted the difficulty of aligning scaffold degradation with the rate of extracellular matrix production by encapsulated cells, while also minimizing immune responses.[26,29] Hydrogel swelling, which increases the distance between electrodes and neurons, further compromises BNI efficacy.[14,26,30] Additionally, cell-created pores may exacerbate hydrogel swelling.[31]

The challenges extend to the implantation process, where mechanical forces can compromise scaffold integrity. Reports of hydrogel detachment or delamination due to shear stress have underscored the need for enhanced surface adhesion techniques to mitigate these issues.[15,26,33]

Over time, changes at the electrode-tissue interface can lead to increased impedance, impairing recording quality. Rochford et al. emphasized the need for robust long-term control strategies to ensure interface stability and maintain implant functionality.[35]

*Cell Viability and Differentiation*: Maintaining cell viability during and after implantation remains a significant challenge for BNIs. Azemi et al. identified difficulties in ensuring that neural progenitor cells adhere to the implant and remain viable throughout the implantation process.[38] Long-term cell survival, particularly in degenerative environments, has been identified as a critical concern.[21] The potential for apoptotic cells and cellular debris to act as toxic factors that impair the function of healthy cells has also been discussed.[17,31] In addition, reviews have emphasized the importance of maintaining long-term cell viability, preventing excessive proliferation, and ensuring the formation of appropriate synaptic connections.[15]

*Synaptic Integration and Connectivity*: Achieving and monitoring proper synaptic integration, as well as promoting directional neurite outgrowth, are essential for the success BNIs.[17] Challenges related to promoting vectorial outgrowth and ensuring targeted synaptic connections have been discussed in several reviews.[22,36] Wise et al. reported that ectopic re-sprouting of SGNs in treated cochleae could reduce the specificity of electrical stimulation, thereby complicating cochlear implant performance.[39] Adewole et al. further highlighted the importance of attaining optimal synaptic density and specificity to avoid unintended neural interactions that could disrupt network functions.[32]

*Electrochemical and Functional Challenges*: Electrochemical stimulation, commonly used to assess implant functionality, may inadvertently affect biological components and compromise cell survival and function.[15] They also highlighted its potential impact on tissue-engineered constructs and surrounding host tissues. Similarly, Goding et al. identified electrochemical damage incurred during electrode fabrication as a factor that can impair overall implant performance.[31]

Additional challenges arise in signal recording, particularly for bidirectional living electrodes. Recording neuronal signals through multiple synapses presents challenges in isolating specific neuronal activity.[36]

*Risk of Tumorigenesis and Safety Concerns*: The inherent proliferative capacity of stem cells increase the risk of tumorigenesis. Controlling cell proliferation without compromising regenerative potential remains a critical concern.[16,21] Programmed cell death has been proposed as a potential solution to eliminate malfunctioning cells and prevent uncontrolled growth.[21]

In addition, the removal of malfunctioning BNIs poses significant challenges due to their integration with host tissues. Procedures involving implanted electrodes can pose a risk of damaging surrounding tissues, which complicates their safe replacement.[21]

*Clinical Biofabrication and Manufacturing Challenges*: The clinical bio-fabrication of BNIs presents several translational challenges. Maintaining sufficient biomass, ensuring consistent quality, and conducting comprehensive safety monitoring remain key challenges for clinical translation.[36] Although autol-


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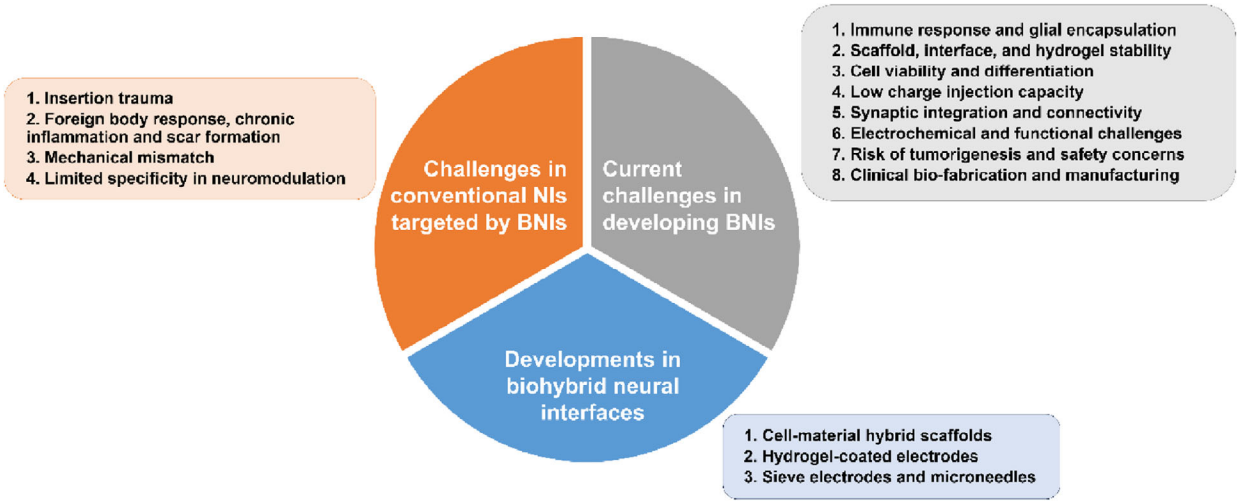


**Figure 2.** Findings from the selected publication.

ogous, stem cell-derived neurons hold significant promise, they are costly and more difficult to validate compared to standardized allogenic constructs, complicating quality control and regulatory approval. The development of personalized electrodes further exacerbate these challenges by introducing variability that hinder standardization.[36]

## 4. Discussion

Biohybrid neural interfaces represent a paradigm shift in interfacing strategies by combining synthetic materials with living biological elements. This review identifies a range of persistent limitations in conventional neural interfaces and highlights how biohybrid strategies offer tailored solutions to overcome these barriers (**Figure 2**). However, the field remains challenged by biological integration, mechanical instability, and translational feasibility. Here, we synthesize key findings across the reviewed literature to map core engineering and biological challenges to emerging solutions and suggest future directions.

### 4.1. Immune Response and Foreign Body Reaction

The glial encapsulation and chronic inflammation observed in traditional NIs undermine long-term function by increasing impedance and reducing signal fidelity. BNIs offer several mechanisms to mitigate these effects. Studies incorporating neural stem cells or supportive glial cells into hydrogel matrices show reduced astrocytic activation in early implantation stages.[28,40] Notably, combining NSCs with immunomodulatory scaffolds (e.g., alginate or fibrin) facilitates localized immune tolerance and neural integration.[26] This immunoregulatory capacity remains highly context-dependent, and variability in host responses underscores the need for personalized immunoengineering approaches.[41]

### 4.2. Mechanical Mismatch

The transition from rigid implants to mechanically compliant materials, such as fibrin, PVA, and composite hydrogels, reflects a significant advance. Studies by De Faveri et al., Green et al., and Cullen et al. demonstrate how mechanical compliance reduces tissue strain, minimizing inflammatory responses.[26,29,42] However, issues such as hydrogel swelling can paradoxically increase electrode-neuron distances, thereby negating electrical efficacy.[15,26] This has led to the development of controlled degradation hydrogels whose breakdown kinetics align with extracellular matrix deposition. Future work must integrate mechanoresponsive elements and swelling-resistant architectures to ensure stable proximity and biofunctionality.[43]

### 4.3. Neural Specificity and Synaptic Integration

Conventional NIs stimulate bulk neural populations, limiting therapeutic precision. BNIs enable targeted synaptic formation through directional neurite outgrowth, guided by scaffold patterning or neurotrophic gradients.[22,36] However, uncontrolled sprouting such as that observed by Wise et al. in cochlear models can degrade signal specificity.[39] There is a need to refine spatial guidance systems using microtopography and bioactive cue gradients that enforce selective axonal pathfinding.[44] Moreover, glial co-culture has emerged as a facilitator of synaptic maturation and homeostasis, but the long-term effects on circuit remodeling remain poorly understood. Functional readouts that track synaptic fidelity in vivo are crucial to advancing this frontier.

### 4.4. Cell Viability and Host Integration

Cellular survival postimplantation is undermined by inflammatory stress, nutrient deficiency, and detachment from scaffolds. The vulnerability of encapsulated cells in hostile environments has been frequently noted.[17,38] Strategies such as glial cell inclusion, apoptotic debris clearance, and preconditioning of constructs aim to enhance resilience. One underexplored but promising approach is the pre-vascularization of scaffolds via endothelial co-culture or inducible angiogenic gene expression, which has shown efficacy in organoid systems.[45,46] This vascular


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**Table 3.** Latest reviews highlighting emerging developments in biohybrid neural interfaces (2025).

| Author(s) | Year of publication | Title of the study | Journal name | Relevance to BNI |
|---|---|---|---|---|
| Boufidis et al.[13] | 2025 | Bio-inspired electronics: Soft, biohybrid, and "living" neural interfaces | Nature Communications | Comprehensive review of biohybrid soft electronics integrating living components with neural devices |
| Carnicer-Lombarte et al.[18] | 2025 | The Future of Biohybrid Regenerative Bioelectronics | Advanced Materials | Discusses next-generation regenerative bioelectronics with BNI relevance |
| Salehi Moghaddam et al.[41] | 2025 | Engineering the Immune Response to Biomaterials | Advanced Science | Explores strategies to mitigate immune response to implanted materials, crucial to BNI integration |
| Wu et al.[43] | 2025 | Recent Progress of Soft and Bioactive Materials in Flexible Bioelectronics | Cyborg and Bionic Systems | Reviews innovations in soft, bioactive materials for flexible bioelectronics enabling stable biological integration |

integration may be essential for translating BNIs to larger animal models and human applications.

### 4.5. Safety and Tumorigenicity

The use of pluripotent or progenitor cells poses a recognized risk of tumorigenesis. Programmed cell death systems or suicide gene cassettes have been proposed as strategies to selectively eliminate overproliferating cells.[16,21] Additionally, predifferentiating cells toward neural lineages and using small molecule inhibitors have shown promise in limiting proliferative escape. Future biohybrid constructs should incorporate multitiered safety mechanisms, possibly including chemical, genetic, and immune-based containment systems.

### 4.6. Clinical Translation

Despite encouraging preclinical results, clinical translation remains limited. Only few works have reached human trials, while most others remain confined to rodent or in vitro models. Key barriers include the scalability of cell sourcing, reproducibility of scaffold-cell integration, and compliance with good manufacturing practice and regulatory frameworks.[13] Autologous constructs, while immunologically advantageous, are costly and time-intensive. Allogeneic or off-the-shelf cellular constructs offer scalability but raise immune compatibility concerns. Future efforts should prioritize standardized modular designs, robust bioprocessing pipelines, and coordinated regulatory engagement early in development.

### 4.7. Limitations of Literature Scope

As the field continues to progress, several recent articles have presented or synthesized emerging evidence that complements and extends the core findings of this review.[13,18,41,43] Although these references were not part of the formal literature collection phase, they offer valuable perspectives on the current trajectory of biohybrid neural interface research. For clarity and transparency, they are summarized in **Table 3** to contextualize ongoing developments and reinforce the relevance of the challenges and solutions discussed herein.

## 5. Conclusion

BNIs hold the potential to redefine neuroprosthetic and neuromodulation therapies by enabling biologically harmonized integration with the nervous system. However, engineering, biological, and translational challenges remain significant. Progress will require interdisciplinary approaches that merge materials science, synthetic biology, and neural engineering. Moreover, real-time monitoring tools and functional validation metrics must be developed to track integration over time. As the field moves forward, collaborative consortia and shared data standards will be essential in bridging the gap from bench to bedside.

## Acknowledgements

This work was supported in part by Tri-Service General Hospital, National Defense Medical Center. M.B. is supported by the National Institute for Health and Care Research (NIHR) Cambridge Biomedical Research Centre (NIHR203312*). The views expressed are those of the authors and not necessarily those of the NIHR or the Department of Health and Social Care.

## Conflict of Interest

The authors declare no conflict of interest.

## Data Availability Statement

The original contributions presented in this study are included in the article/Supporting Information. Further inquiries can be directed to the corresponding authors.


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## Keywords

biocompatible electrodes, biohybrid neural interfaces, cell-seeded electrodes, hydrogel neuronal coating, living electrodes

Received: April 14, 2025
Revised: September 24, 2025
Published online:

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